ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a reversible chemical change	Hydrated and anhydrous forms.
2	**(c)** formation of a new substance	New substance.
3	**(b)** store in dry place or oil them	Keeps moisture away.
4	**(a)** copper sulphate	Copper deposits.
5	**(d)** lime water test	Test gas.
6	**(c)** apply oil and keep dry	Oiling.
7	**(c)** carbon dioxide	Lime water test.
8	**(d)** unburnt carbon, a chemical change	New substance.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Only the state changes from solid to liquid, so R explains why it is physical.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but being a metal does not explain the need for oxygen and water.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Iron combines with oxygen and water (1); Rust has extra mass (1).	1+1
12	Burning forms ash and gases (1); Tearing only changes size and shape (1).	1+1
13	Takes a long time (1); Small changes each day (1).	1+1
14	No new substance in melting (1); Magnesium oxide forms (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A: no rust (no moisture) (1); B: no rust (no air) (1); C: rusts (air and water) (1).	1+1+1
16	Salt and sand do not react (1); Iron and sulphur react (1); New compound formed (1).	1+1+1
17	Burning: chemical (1); Light, sound, heat (1); Smoke and gases (1).	1+1+1
	OR: Chemical (1); New substance slaked lime (1); Heat released (1).	1+1+1
18	Burning candle (1); Melting is physical (1); Burning vapour is chemical (1).	1+1+1
	OR: Combines with iron (1); Combines with fuel (1); Oxides form (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Crystallisation (1). (ii) Physical change (1). (iii) More copper sulphate dissolves in hot water, so the solution becomes saturated and crystals form on cooling (1). (iv) Cool the solution slowly and keep it undisturbed (1), so the particles have time to arrange themselves and grow into larger crystals (1).	1+1+1+2
20	LPG burns: chemical (1); Milk warms: physical (1); Milk boils: physical (1); Milk burns at bottom: chemical (1); Black deposit is carbon (1).	1+1+1+1+1
	OR: Washing dissolves dirt: physical (1); Dry cleaning solvent (1); Silver polish: chemical (1); Rust removal by acid: chemical (1); Choose method (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that every change of state is chemical; melting, freezing and boiling are **physical** changes.
- Calling dissolving of sugar or salt in water a chemical change; the substance can be recovered by evaporation.
- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** a chemical change	Energy released.
2	**(c)** silver reacts with sulphur in air	Alloy resists.
3	**(c)** a new black solid not attracted by a magnet	New compound.
4	**(a)** melting of ice	Heat taken in.
5	**(b)** to resist rusting	Longer life.
6	**(d)** rust is brittle and flakes off	Weakens.
7	**(d)** burning of LPG	Combustion.
8	**(c)** new products carbon dioxide and water	Combustion.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Melting is physical and burning is chemical, which R states and so explains A.
10	**(d)** A is false but R is true.	No new substance is formed, so it is a physical change and A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Boiling removes dissolved air (1); So only water, not air, is tested (1).	1+1
12	Higher temperature (1); Steam speeds the change (1).	1+1
13	Water of crystallisation lost can be regained (1); Ash cannot become paper (1).	1+1
14	It does not relight a glowing splinter (1); It turns lime water milky (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Warm place speeds (1); Cold slows (1); New substances formed (1).	1+1+1
16	Both need oxygen (1); Burning is fast, rusting slow (1); Burning gives heat and light (1).	1+1+1
17	Chlorophyll breaks down (1); Other pigments visible (1); Chemical change (1).	1+1+1
	OR: Physical (1); Same substance (1); Energy used for change of state (1).	1+1+1
18	Reacts with chalky carbonate (1); Releases carbon dioxide (1); Chemical change (1).	1+1+1
	OR: Cooking (1); Burning (1); New substances formed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Rust (hydrated iron oxide) (1). (ii) Oxygen (air) and moisture (water) (1). (iii) Chemical change (1). (iv) Painting and galvanisation (zinc coating) (1); both keep air and water away from the iron surface (1).	1+1+1+2
20	Bridges and rails damaged (1); Cost of repairs (1); Paint and galvanise (1); Use stainless steel (1); Regular maintenance (1).	1+1+1+1+1
	OR: Shaping: physical (1); Drying: physical (1); Firing in kiln: chemical (1); Hard pot formed (1); Cannot return to clay (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** gas bubbles and new substances form	Yeast action.
2	**(c)** a chemical change	New substances such as carbon form.
3	**(c)** a chemical reaction releases heat	Slaking.
4	**(d)** burning of a candle	Mix.
5	**(a)** the batter rises and changes colour	Gas and colour change.
6	**(b)** slaked lime, a new substance, forms	New product.
7	**(c)** reaction with air oxygen is slowed	Isolation from air.
8	**(c)** dissolving salt in water	Only dissolving is physical.
9	**(c)** A is true but R is false.	A is true, but the coating used in galvanisation is zinc, so R is false.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Carbon dioxide turns lime water milky, so R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Surface becomes iron oxide (1); Dull flaky layer forms (1).	1+1
12	Yeast produces gas (1); Chemical change from fermentation (1).	1+1
13	Forms silver sulphide (1); Black coating (1).	1+1
14	New substances like carbon form (1); Chemical change (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Zinc coating (1); Alloy with chromium (1); Resist corrosion (1).	1+1+1
16	Crystals contain water (1); Water is released as vapour (1); Condenses (1).	1+1+1
17	Add water to quicklime (1); Hissing and heat (1); Touch the beaker carefully / use thermometer (1).	1+1+1
	OR: Substances in the air react with silver (1); A black coating forms (1); It is a chemical change (1).	1+1+1
18	Milk to curd: chemical (1); Churning: physical (1); Butter separated (1).	1+1+1
	OR: Glucose reacts with oxygen (1); Energy released (1); Carbon dioxide formed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Physical change (1). (ii) Chemical change (1). (iii) Physical change (1). (iv) Chemical change (1): a new substance (curd) with a different taste and smell is formed, and milk cannot be obtained back from curd (1).	1+1+1+2
20	Microbes act on batter (1); Gas bubbles (1); Dough rises (1); Curd forms (1); New substances (1).	1+1+1+1+1
	OR: List steps for each dish (1); Mark physical steps (1); Mark chemical steps (1); Table of results (1); Conclusion (1).	1+1+1+1+1

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